

Annie Guo

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Education

University of Waterloo

Sep 2022 - May 2026

Bachelor of Computer Science

Waterloo, ON

- Specializing in Artificial Intelligence (GPA: 3.7/4.0)
- Coursework: Object-Oriented Programming, Data Structures, Algorithms, Compilers, Operating Systems
- Extracurriculars: UW Blueprint Technical Lead, Data Science Club Project Developer, Varsity Figure Skating

Experience

Amazon Web Services

May 2025 - Present

Software Engineer Intern

Vancouver, BC

- Optimized the horizontal scaling process for internal EKS infrastructure, reducing cluster provisioning time by ~1 week by introducing actionable properties into cluster data models and refactoring CI/CD pipeline workflows
- Collaborated on the development of a CDK constructs library to help users launch production-ready EKS environments with CloudFormation, following cell based architecture and Kubernetes best practices

Ford Pro

Jan 2025 - Apr 2025

Software Engineer Intern

Oakville, ON

- Created a data migration pipeline, transferring 100,000+ company account records into a PostgreSQL database, using GraphQL to handle API interactions and Google Cloud Run for serverless execution
- Implemented an user enrollment endpoint within a Spring Boot microservice that enables companies to seamlessly register their users for Ford products, using Java and Azure IAM
- Built interactive UI components for the dashboard, providing financing analytics and currently serving 2,000+ daily active users, with React, TypeScript, and SCSS

Descartes Systems Group

May 2024 - Aug 2024

Software Engineer Intern

Waterloo, ON

- Migrated critical company applications from legacy frameworks to .NET Core through modularizing programs and refactoring 10,000+ lines of code to adopt MVC architecture
- Engineered a geocoding service in C# and SQL that supports address validation across 18 countries by integrating with multiple third-party APIs and caching frequently searched addresses in-memory to optimize performance

Eon Media

Sep 2023 - Dec 2023

Software Engineer Intern

Toronto, ON

- Accelerated a video encoding pipeline to process 20× more videos daily and convert 12,000+ hours of MP4 content into HLS playlists by leveraging AWS Lambda to execute FFmpeg commands in parallel
- Improved brand recognition performance within an AI video processing pipeline by 25% through image augmentation and text preprocessing using OpenCV, Tesseract OCR, and SciPy
- Analyzed cost inefficiencies across multiple AWS resources, reducing monthly cloud expenses by over \$3,000 through S3 Glacier storage lifecycle policies, EC2 spot instance migration, and Lambda optimizations
- Deployed a Slack chatbot to deliver real-time notifications on active pipelines and control pipeline operations via chat commands, automating workflows and cutting manual actions by half using Python and Slack API
- Designed an autocompletion system using Elasticsearch to streamline brand name lookups
- Built Flask API endpoints enabling frontend teams to perform CRUDL operations on video metadata

Skills

Languages: Python, Go, Javascript, Typescript, C++, Java, SQL, HTML, CSS

Frameworks & Tools: React, .NET Core, Spring Boot, Docker, Kubernetes, OpenCV, Tensorflow, Git, Tailwind

Database & Cloud: AWS, Azure, Google Cloud, MS SQL, PostgreSQL, MongoDB